

CA2 – CFGs and PDAs

1. The language $L1 = \{w\#w \mid w \text{ is in } \{0, 1\}^*\}$ is not a context-free language (CFL).

2. Justify, using what CFGs are capable of, why $L1$ is not a CFL; 
(1 Ponto)

3. Justify, using what PDAs are capable of, why $L1$ is not a CFL; 
(1 Ponto)

4. As you know, the complement of a CFL may not be a CFL. Show that the complement of $L1$ is a CFL by presenting a possible CFG for the complement of $L1$; 
(7 Pontos)

5. Instead of a stack such as in the PDAs, one can think about other FA extensions. Select the kind of structures that could be used to extend an FA and make it able to implement $L1$: 
(1 Ponto)

☐ queue

☐ 2 stacks

☐ none of the other options;